

Senzo Ncekana

AI ENGINEER — AGENTIC AI, LLMS & BACKEND SYSTEMS

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PROFESSIONAL SUMMARY

AI engineer who builds practical, production-minded AI systems — agentic assistants, RAG pipelines, and the Python backends behind them. Hands-on with LangChain agents and tools, OpenAI and Google Gemini LLMs, vector search (Pinecone, FAISS, pgvector), and backend engineering with FastAPI, Django, REST APIs, PostgreSQL, Docker, and CI/CD. Recently built a multi-channel agentic AI assistant in a part-time engineering role. 5+ years in regulated healthcare instilled an enterprise mindset: data integrity, compliance, documentation, and reliability. A fast learner focused on enterprise AI solutions that solve real-world business challenges.

TECHNICAL SKILLS

AI Engineering: LLMs (OpenAI, Google Gemini), LangChain agents & tools, RAG / vector search (Pinecone, FAISS, pgvector), prompt design, NLP (spaCy), applied machine learning (supervised & unsupervised, scikit-learn, XGBoost), neural networks, model evaluation, testing & validation of AI outputs, responsible AI awareness.

Backend Engineering: Python, FastAPI, Django, REST API design & integration, relational database design (PostgreSQL, SQLite), SQL, Docker, CI/CD, Git/GitHub, n8n workflow automation, cloud-hosted services (Pinecone).

Data & Analytics: pandas, ETL / data preparation, exploratory data analysis (EDA), statistical analysis, anomaly & exception detection, Power BI, Tableau, Excel, dashboards, reporting.

Soft Skills: Problem-solving · analytical thinking · collaboration across technical and non-technical teams · attention to detail · managing multiple priorities · comfortable with ambiguity / Agile · fast learner.

WORK EXPERIENCE

AI/ML Engineer (Part-Time) — Datainno Analytics

Feb 2026 – Jul 2026

- Built an agentic AI concierge for hotel operations using LangChain (tools/agents), OpenAI, and RAG over a Pinecone vector database, grounding responses in business data to reduce hallucinations.
- Engineered LangChain tools that call backend APIs to check availability, process bookings, update room inventory, and log guest interactions — automating end-to-end reservation workflows.
- Integrated the assistant across WhatsApp, Telegram, and web channels with intent classification, entity extraction, and multi-turn context retention; tested, validated, and monitored AI outputs through rollout.
- Built an analytics layer surfacing guest-satisfaction sentiment, chat-to-booking conversion, upsell revenue, and peak-demand trends for management decision-making.

Pharmacy Technician — Nelson Mandela Children's Hospital

Jul 2023 – Present

- Maintain accurate electronic records and inventory data in a regulated environment, ensuring data integrity for reporting and audits.
- Analyse stock and consumption data to detect exceptions and anomalies, supporting formulary and stock-control decisions.
- Identify and implement workflow improvements in a high-volume dispensary; collaborate with clinical, operational, and administrative teams.

Pharmacy Technician — Local Choice Pelican Pharmacy

Jan 2022 – Jun 2023

- Ensured accurate data entry, record-keeping, and data integrity across dispensary operations under strict regulatory compliance.
- Monitored stock trends to flag exceptions and reduced dispensing errors through process improvements.

Pharmacy Technician Intern — Prince Mshiyeni Memorial Hospital

Apr 2021 – Nov 2021

- Monitored inventory levels and prepared reporting to support audits and regulatory compliance; documented patient and stock data accurately.

Lab Assistant (Manufacturing) — Nelson Mandela University

Jun 2019 – Nov 2019

- Collected, recorded, and analysed production data to identify efficiency improvements; performed quality-control checks against safety and quality standards in regulated manufacturing.

PROJECTS

PharmaGuard — Pharmaceutical Similarity Detection & Compliance Platform

End-to-end AI platform detecting look-alike/sound-alike (LASA) risks in drug names and packaging: multi-agent pipeline with RAG and OCR/computer vision behind a FastAPI backend and Next.js frontend; Dockerised full-stack app with CI/CD. **Tech:** Python, FastAPI, Next.js, Google Gemini, LangChain, FAISS, PaddleOCR, PostgreSQL, Docker.

Customer Churn — Predictive Risk-Scoring Pipeline

End-to-end machine learning pipeline (EDA, data cleaning, feature engineering, model comparison, ROC-AUC/F1) scoring customer churn risk to support targeted, data-driven retention decisions. **Tech:** Python, pandas, scikit-learn, XGBoost, Matplotlib, Seaborn.

Amazon Review Sentiment Analysis — NLP on Unstructured Data at Scale

NLP pipeline over 28,000+ product reviews (preprocessing, polarity classification, sentiment scoring, similarity analysis) with reproducible, documented workflows. **Tech:** Python, spaCy, SpacyTextBlob, Jupyter.

US Crime Statistics — Pattern & Anomaly Detection

Segmented a full state-level dataset and detected patterns and outliers using PCA with K-means and hierarchical clustering, presenting statistical interpretation and visualisation for non-technical readers. **Tech:** Python, pandas, scikit-learn, SciPy, Matplotlib, Seaborn.

Shelf Track — Relational Database & Inventory Management System

Relational inventory database with full CRUD and search, prioritising data integrity and usability. **Tech:** Python, SQLite, OOP, database design.

EDUCATION & CERTIFICATIONS

AI Engineering Bootcamp — HyperionDev

Feb 2025 – Jan 2026

- 12-month intensive programme: data ingestion and ETL, supervised and unsupervised machine learning, neural networks, NLP, SQL and relational database design, and full-stack Django development, using Git/GitHub throughout.
- Built end-to-end, reproducible workflows for structured and unstructured data, with dashboards turning data into decision-ready insight.

n8n Workflow Automation — Certification

- Workflow automation end-to-end: triggers, core nodes and expressions; REST API and webhook integration; transforming and routing data between systems; scheduling and error handling.
- Applies directly to data pipeline work — moving, reshaping and validating data between business systems without writing custom integration code.

Machine Learning Fundamentals — Udemy

2025 – 2026

- Core machine learning workflow: supervised and unsupervised methods — regression, classification and clustering — with train/test splitting, cross-validation, feature engineering, and managing overfitting.
- Model evaluation and comparison using precision, recall, F1 and ROC-AUC, applied directly in my churn-prediction and clustering projects.

Diploma in Pharmacy Technical Support — Nelson Mandela University

2017 – 2019

PROFESSIONAL INTERESTS

Passionate about enterprise AI, large language models, and agentic systems. Build open-source and personal side projects to explore new frameworks, sharpen engineering skills, and turn ideas into working, scalable systems — and actively follow responsible AI practice, including the risks and opportunities AI presents to business.